

Practical 1

1. Study of Network-Related Commands (Windows)

a. **Hostname** : Shows computer's hostname.

```
C:\Users\Dell>hostname
Sahil-Lodwal

C:\Users\Dell>
```

b. **systeminfo** :

- Shows detailed system info including network configs.
- Includes:
 - OS version
 - Hotfixes
 - Network card configuration

```
C:\Users\Dell>systeminfo

Host Name:                SAHIL-LODWA
OS Name:                  Microsoft Windows 10 Pro
OS Version:               18.0.19045 N/A Build 19045
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:         Dell
Registered Organization:
Product ID:                00330-50864-56976-AA0EM
Original Install Date:     04-04-2023, 08:13:37
System Boot Time:         03-04-2025, 16:46:32
System Manufacturer:      Dell Inc.
System Model:              Latitude 3480
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 142 Stepping 9 GenuineIntel ~2701 Mhz
BIOS Version:              Dell Inc. 1.24.0, 11-04-2023
Windows Directory:        C:\Windows
System Directory:          C:\Windows\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:              en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:     7,901 MB
Available Physical Memory: 2,951 MB
Virtual Memory: Max Size: 9,949 MB
Virtual Memory: Available: 3,141 MB
Virtual Memory: In Use:    6,808 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:              \\SAHIL-LODWA
Hotfix(s):                 32 Hotfix(s) Installed.
                           [01]: KB50590576
                           [02]: KB5038041
                           [03]: KB5007401
                           [04]: KB5011048
                           [27]: KB5043130
                           [28]: KB5046823
                           [29]: KB5050388
                           [30]: KB5050111
                           [31]: KB5052916
                           [32]: KB5054682
Network Card(s):           5 NIC(s) Installed.
                           [01]: Qualcomm QCA61x4A 802.11ac Wireless Adapter
                               Connection Name: Wi-Fi
                               DHCP Enabled: Yes
                               DHCP Server: 192.168.17.37
                               IP address(es)
                               [01]: 192.168.17.184
                               [02]: fe80::5dc1:667e:8724:1b8
                               [03]: 2402:3a80:4558:957c:ed09:c432:e69c:f4a
                               [04]: 2402:3a80:4558:957c:ea85:8ecd:91d9:69b3
                           [02]: Realtek PCIe GBE Family Controller
                               Connection Name: Ethernet
                               Status: Media disconnected
                           [03]: Bluetooth Device (Personal Area Network)
                               Connection Name: Bluetooth Network Connection
                               Status: Media disconnected
                           [04]: Hyper-V Virtual Ethernet Adapter
                               Connection Name: vEthernet (Default Switch)
                               DHCP Enabled: No
                               IP address(es)
                               [01]: 172.17.48.1
                               [02]: fe80::e568:41ac:d391:efa8
                           [05]: TAP-Windows Adapter V9
                               Connection Name: Local Area Connection
                               Status: Media disconnected
Hyper-V Requirements:      A hypervisor has been detected. Features required for Hyper-V will not be displayed.
```

```
[05]: KB5012170
[06]: KB5015684
[07]: KB5055518
[08]: KB5014032
[09]: KB5014671
[10]: KB5020372
[11]: KB5022924
[12]: KB5023794
[13]: KB5025315
[14]: KB5026879
[15]: KB5028318
[16]: KB5029709
[17]: KB5031539
[18]: KB5032392
[19]: KB5032907
[20]: KB5034224
[21]: KB5037018
[22]: KB5037240
[23]: KB5039336
[24]: KB5041570
[25]: KB5041581
[26]: KB5043935
[27]: KB5043130
[28]: KB5046823
[29]: KB5050388
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    DHCP Server: 192.168.17.37
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    [01]: 192.168.17.184
    [02]: fe80::5dc1:667e:8724:1b8
    [03]: 2402:3a80:4558:957c:ed09:c432:e69c:f4a
    [04]: 2402:3a80:4558:957c:ea85:8ecd:91d9:69b3
[02]: Realtek PCIe GBE Family Controller
```

```

C:\Users\Dell>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix . : 
Ethernet adapter Ethernet (Default Switch):

   Connection-specific DNS Suffix . : 
   Link-local IPv6 Address . . . . . : fe80::5d01:607e:8724:10ba%1
   IPv4 Address. . . . . : 192.168.17.184
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : 192.168.17.1

Unknown adapter Local Area Connection:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix . : 
Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix . : 
Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix . : 
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix . : 
   IPv6 Address. . . . . : 2400::6800:4009:80b::200e
   Temporary IPv6 Address . . . . . : 2402::3a00:4508:957c:1ea0:80c0:91d9:69b3
   Link-local IPv6 Address . . . . . : fe80::5d01:607e:8724:10ba%1
   IPv4 Address. . . . . : 192.168.17.184
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : fe80::17ea:effe:febe:9f00%2
   192.168.17.37

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix . : 

```

c. **Ipconfig (IP Configuration)** : Displays and manages **IP configuration** of your system.

d. **Ping** : Tests **connectivity** with another device or server.

```

C:\Users\Dell>ping google.com

Pinging google.com [2404:6800:4009:80b::200e] with 32 bytes of data:
Reply from 2404:6800:4009:80b::200e: time=54ms
Reply from 2404:6800:4009:80b::200e: time=75ms
Reply from 2404:6800:4009:80b::200e: time=127ms
Reply from 2404:6800:4009:80b::200e: time=112ms

Ping statistics for 2404:6800:4009:80b::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 54ms, Maximum = 127ms, Average = 92ms

```

```

C:\Users\Dell>nslookup
Default Server: UnKnown
Address: 192.168.17.37

> google.com
Server: UnKnown
Address: 192.168.17.37

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4009:80b::200e
          142.250.195.14

> exit

```

e. **nslookup** : used to **query DNS servers** to find the IP address of a domain or the domain name of an IP.

f. **tracert** : shows the **path (hops)** packets take to reach a destination and measures delay at each hop

```

C:\Users\Dell>tracert google.com

Tracing route to google.com [142.250.182.46]
over a maximum of 30 hops:

  0  0 ms  0 ms  0 ms  192.168.0.1
  1  2 ms  4 ms  5 ms  ec2-100-62-0-1.compute-1.amazonaws.com [100.62.0.1]
  2  4 ms  3 ms  4 ms  103.182.134.45
  3  5 ms  *  7 ms  72.14.209.190
  4  7 ms  5 ms  6 ms  142.251.77.23
  5  5 ms  6 ms  4 ms  142.251.76.169
  6  4 ms  6 ms  5 ms  maa05s19-in-f14.1e100.net [142.250.182.46]
  7  5 ms  6 ms  7 ms 

Trace complete.

```